

Problem

A second-order active circuit has the transfer function

$$H(s) = \frac{1}{s^2 + s(25 + \alpha) + 25}$$

Find the range of the values of α for which the circuit is stable. What is the value of α that will cause oscillation?

Step-by-step solution

Step 1 of 1

Given second-order active circuit has the transfer function.

$$H(s) = \frac{1}{s^2 + s(25 + \alpha) + 25}$$

This has poles at

$$P^2 + bp + C = 0$$

Where $b = 25 + \alpha, c = 25$

$$P_{1,2} = \frac{-b \pm \sqrt{b^2 - 4C}}{2}$$

$$P_{1,2} = \frac{-(25 + \alpha) \pm \sqrt{(25 + \alpha)^2 - 4(25)}}{2}$$

For the circuit to be stable, the poles must be located in the left half of the s-plane

Which implies

$$\Rightarrow b > 0$$

Applying these conditions to $H(s)$, then the circuit to be stable,

$$25 + \alpha > 0 \text{ or}$$

$$\therefore \alpha > -25$$